ACTOR-CRITIC METHODS

INTERMEDIATE DEEP LEARNING

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PREVIOUSLY…

POLICY GRADIENT METHODS AND REINFORCE

PREVIOUSLY …

POLICY GRADIENT METHODS AND REINFORCE

- Policy gradient methods are a class of reinforcement learning algorithms that directly optimize the policy
- Estimate the gradient of expected return with respect to the policy parameters and updating them in the direction that increases expected reward.

$$\nabla_\theta J(\pi_\theta) = E_{\tau \sim \pi_\theta} \left[\sum_{t=0}^{T} R_t(\tau) \, \nabla_\theta \log \pi_\theta \left(a_t | s_t \right) \right]$$

PREVIOUSLY …

POLICY GRADIENT METHODS AND REINFORCE

- REINFORCE is a Monte Carlo-based policy gradient algorithm
- But it has some drawbacks :
  - High variance
  - Credit assignment problem
  - Inefficient updates (only after an episode finishes)



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POLICY GRADIENT METHODS AND REINFORCE

- REINFORCE is a Monte Carlo-based policy gradient algorithm
- But it has some drawbacks :
  - High variance
  - Credit assignment problem
  - Inefficient updates (only after an episode finishes)
- What if we can better evaluate the influence of action?



INTRODUCTION

ACTOR-CRITIC METHODS

ACTOR-CRITIC METHODS



Actor



Critic

INTRODUCTION

ACTOR-CRITIC METHODS







Actor



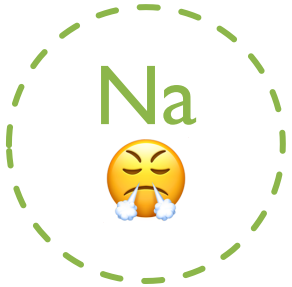
Critic

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Actor

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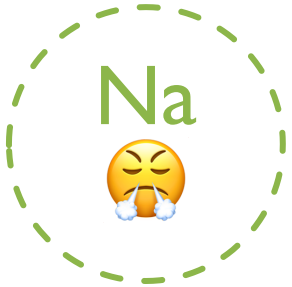






Actor







Critic

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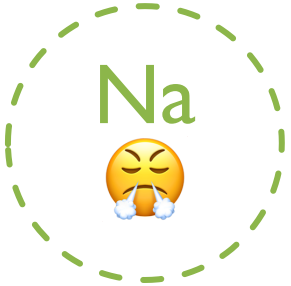






Actor







Critic

ACTOR-CRITIC METHODS

Policy function







Actor

Actions

Yess

😍

Na

😤

Values

Value function



Critic

INTRODUCTION

ACTOR-CRITIC METHODS

- A learned reinforcement signal can be more informative for a policy than the raw rewards provided by the environment
- A common option is the advantage function:

$$A_\pi(s,a) = Q_\pi(s,a) - V_\pi(s)$$

- Where $A_\pi(s,a)$: the advantage,
- $Q_\pi(s,a)$: the expected return of taking action a in state s
- $V_\pi(s)$: the value (~expected return) of state s when following the current policy

ADVANTAGE ACTOR-CRITIC (A2C)

ADVANTAGE – N-STEPS RETURNS – GENERALIZED ADVANTAGE ESTIMATION (GAE)

ADVANTAGE FUNCTION

$$A_\pi(s,a) = Q_\pi(s,a) - V_\pi(s)$$

- Intuition: $A_\pi(s,a)$ measures how much better or worse an action is compared to the average action chosen by the policy in that state.
- If an action is worse than the average but still receives a positive return?
  - E.g.

$$Q_\pi(s,a) = 90, \quad V_\pi(s) = 100, \quad A_\pi(s,a) = -10$$

or

$$Q_\pi(s,a) = -90, \quad V_\pi(s) = -100, \quad A_\pi(s,a) = 10$$

- If the policy is already good?

ADVANTAGE ACTOR-CRITIC (A2C)

ADVANTAGE FUNCTION

$$A_\pi(s,a) = Q_\pi(s,a) - V_\pi(s)$$

- It's a "baseline-adjusted" estimate of an action's influence → helps reduce variance in learning.
- It can capture long-term effects of the current action without directly attributing the outcome to previous actions → focuses on what matters now
- But how to approximate this function?

ADVANTAGE FUNCTION

$$A_\pi(s,a) = Q_\pi(s,a) - V_\pi(s)$$

- It's the critic's job to estimate the values used in the advantage function.
- One critic for $Q_\pi(s,a)$ and one for $V_\pi(s)$

$Q_\pi(s,a)$





$V_\pi(s)$

ADVANTAGE FUNCTION

$$A_\pi(s,a) = Q_\pi(s,a) - V_\pi(s)$$

- It's the critic's job to estimate the values used in the advantage function.
- One critic for $Q_\pi(s,a)$ and one for $V_\pi(s)$?
  - Less efficient to train
  - Two estimates should be consistent

$Q_\pi(s,a)$  👍  $V_\pi(s)$

ADVANTAGE ACTOR-CRITIC (A2C)

N-STEP RETURNS

- Typically, we learn just V_π and combine it with reward to estimate Q_π.
  - Q_π is more complex and needs more samples
  - Using Q_π to estimate V_π is more computationally expensive

ADVANTAGE ACTOR-CRITIC (A2C)

N-STEP RETURNS

- The goal is to use $Q_\pi(s,a)$ and $V_\pi(s)$ to approximate return G
- Reminder: $G_t = \sum_{k=t}^{T} \gamma^{k-t} r_k$
- If

$$V_\pi(s_t) = E_\pi[\, G_t \mid s_t \,], \qquad Q_\pi(s_t, a_t) = E_\pi[\, G_t \mid s_t, a_t \,]$$

- Then:

$$Q_\pi(s_t, a_t) = E[\, r_t \mid s_t, a_t \,] + \gamma E_\pi[\, G_{t+1} \mid s_{t+1} \,]$$

$$Q_\pi(s_t, a_t) \approx r_t + \gamma V_\pi(s_{t+1})$$

- Which is a one−step approximation (sounds familiar?)

ADVANTAGE ACTOR-CRITIC (A2C)

N-STEP RETURNS

- We can expand to more steps: each sampled return provides an unbiased sample of the expected return.
- But going too far into the future increases variance
- So we use an n-step return:

$$Q_\pi(s_t, a_t) \approx r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \cdots + \gamma^{n-1} r_{t+n-1} + \gamma^n V_\pi(s_{t+n})$$

- n controls the trade-off between bias and variance
  - When $n = 1$?
  - When $n = T - t$?

N-STEP RETURNS

- Substituting back into

$$A_\pi(s,a) = Q_\pi(s,a) - V_\pi(s)$$

- We obtain:

$$A_\pi^{\text{n-step}}(s_t,a_t) \approx r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \cdots + \gamma^{n-1} r_{t+n-1} + \gamma^n V_\pi(s_{t+n}) - V_\pi(s_t)$$

- Which is the formula of advantage estimated with n-steps return.

ADVANTAGE ACTOR-CRITIC (A2C)

GENERALIZED ADVANTAGE ESTIMATION (GAE)

- With n-step returns, we are already balancing bias and variance by fixing n.
- However,
  - Choosing n is difficult
  - The trade-off is *abrupt or discontinuous* (it changes at a specific step)
- Is there a way to make this balance softer and smoother?

GENERALIZED ADVANTAGE ESTIMATION (GAE)

- Generalized Advantage Estimation (GAE) was proposed as an advantage estimator that provides a continuous trade-off between bias and variance.

$$A_{\pi}^{\mathrm{GAE}}(s_t, a_t) = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}$$

- Where

$$\delta_t = r_t + \gamma V_{\pi}(s_{t+1}) - V_{\pi}(s_t), \qquad \lambda \in [0,1]$$

- Intuitively, each TD error measures how "surprising" a step is, and λ controls how far into the future these surprises still matter

GENERALIZED ADVANTAGE ESTIMATION (GAE)

$$A_{\pi}^{\mathrm{GAE}}(s_t, a_t) = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}$$

- Where

$$\delta_t = r_t + \gamma V_{\pi}(s_{t+1}) - V_{\pi}(s_t), \qquad \lambda \in [0,1]$$

- Where does this λ come from?
- The idea behind GAE is to combine all n-step returns for $n \in \{1,2, \dots\}$…

ADVANTAGE ACTOR-CRITIC (A2C)

GENERALIZED ADVANTAGE ESTIMATION (GAE)

- Idea: GAE can be viewed as an exponentially weighted average (EWA) of all the n-step advantage estimates.
- Given a sequence $(X_1, X_2, \dots, X_t, \dots)$, the exponentially weighted average of X is:

$$Y_t = \lambda Y_{t-1} + (1 - \lambda) X_t$$



ADVANTAGE ACTOR-CRITIC (A2C)

GENERALIZED ADVANTAGE ESTIMATION (GAE)

- If we denote n-step advantage estimate by $A_\pi(n)$:

$$A_\pi(1) = r_t + \gamma V_\pi(s_{t+1}) - V_\pi(s_t),$$

$$A_\pi(2) = r_t + \gamma r_{t+1} + \gamma^2 V_\pi(s_{t+2}) - V_\pi(s_t),$$

$$\dots$$

- Then the EWA of all $A_\pi(n)$ estimates is:

$$(1 - \lambda)\,(A_\pi(1) + \lambda A_\pi(2) + \lambda^2 A_\pi(3) + \cdots)$$

GENERALIZED ADVANTAGE ESTIMATION (GAE)

- Since

$$\delta_t = r_t + \gamma V_\pi(s_{t+1}) - V_\pi(s_t)$$

- we can rewrite $A_\pi(n)$ into:

$$A_\pi(1) = r_t + \gamma V_\pi(s_{t+1}) - V_\pi(s_t) = \delta_t,$$

$$A_\pi(2) = r_t + \gamma r_{t+1} + \gamma^2 V_\pi(s_{t+2}) - V_\pi(s_t) = \delta_t + \gamma\delta_{t+1},$$

$$A_\pi(3) = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \gamma^3 V_\pi(s_{t+3}) - V_\pi(s_t) = \delta_t + \gamma\delta_{t+1} + \gamma^2\delta_{t+2},$$

$$A_\pi(n) = \sum_{k=0}^{n-1} \gamma^k \delta_{t+k}$$

GENERALIZED ADVANTAGE ESTIMATION (GAE)

$$(1-\lambda)(A_\pi(1)+\lambda A_\pi(2)+\lambda^2 A_\pi(3)+\cdots)=(1-\lambda)\sum_{n=1}^{\infty}\lambda^{n-1}A_\pi(n)$$

$$=(1-\lambda)\sum_{n=1}^{\infty}\lambda^{n-1}\sum_{k=0}^{n-1}\gamma^k\delta_{t+k}$$

Combine the two sums

$$=(1-\lambda)\sum_{n=1}^{\infty}\sum_{k=0}^{n-1}\lambda^{n-1}\gamma^k\,\delta_{t+k}$$

Swap the order of summation

$$\{(k,n):\ k\geq 0,\ n\geq k+1\}$$
$$=\{(n,k):\ n\geq 1,\ 0\leq k\leq n-1\}$$

$$=(1-\lambda)\sum_{k=0}^{\infty}\gamma^k\delta_{t+k}\sum_{n=k+1}^{\infty}\lambda^{n-1}$$

$$=(1-\lambda)\sum_{k=0}^{\infty}\gamma^k\delta_{t+k}\left(\frac{\lambda^k}{1-\lambda}\right)=\sum_{k=0}^{\infty}(\gamma\lambda)^k\delta_{t+k}$$

geometric series

ADVANTAGE ACTOR-CRITIC (A2C)

CRITIC TRAINING

- We need to define the loss function for the critic.
- The choice of target can vary, e.g. a simple one-step target

$$V_{\pi}^{\text{target}}(s_t) = r + \gamma V_{\pi,\theta_C}(s_{t+1})$$

- Or we can "recycle" some formulations…

ADVANTAGE ACTOR-CRITIC (A2C)
CRITIC TRAINING

- Intuitively, as the policy converges, V_π approaches Q_π, and the advantage (the "surprise") becomes smaller
- The critic's target is often expressed as

$$V_\pi^{\text{target}}(s_t) = A_\pi(s_t, a_t) + V_{\pi,\theta_C}(s_t).$$

- In practice for n-step return:

$$V_\pi^{\text{target}}(s_t) = r_t + \gamma r_{t+1} + \cdots + \gamma^{n-1} r_{t+n-1} + \gamma^n V_{\pi,\theta_C}(s_{t+n}).$$

- And for GAE:

$$V_\pi^{\text{target}}(s_t) = A_\pi^{\text{GAE}}(s_t, a_t) + V_{\pi,\theta_C}(s_t).$$

ADVANTAGE ACTOR-CRITIC (A2C)

ACTOR TRAINING

- The actor is trained with policy gradient.
- Its update rule is similar to REINFORCE, except that we replace the trajectory return $R_t(\tau)$ with an advantage estimate $A_\pi(s_t, a_t)$ to reduce variance
- REINFORCE:

$$\nabla_\theta J(\pi_\theta) = E_{\tau \sim \pi_\theta}\left[\sum_{t=0}^{T} R_t(\tau)\, \nabla_\theta \log \pi_\theta\,(a_t|s_t)\right]$$

- A2C:

$$\nabla_\theta J(\pi_\theta) = E_{\tau \sim \pi_\theta}\left[\sum_{t=0}^{T} A_\pi(s_t, a_t)\, \nabla_\theta \log \pi_\theta\,(a_t|s_t)\right]$$

ADVANTAGE ACTOR-CRITIC (A2C)

ACTOR TRAINING

- We can implement entropy regularization to encourage exploration:

$$L_{\text{actor}}(\theta_A) = -\log \pi\, \theta_A(a_t|s_t) A_\pi(s_t, a_t) - \beta H_t,$$

- Where

$$H_t = -\sum_a \pi(a|s_t) \log \pi\,(a|s_t)\,, \qquad \beta > 0$$

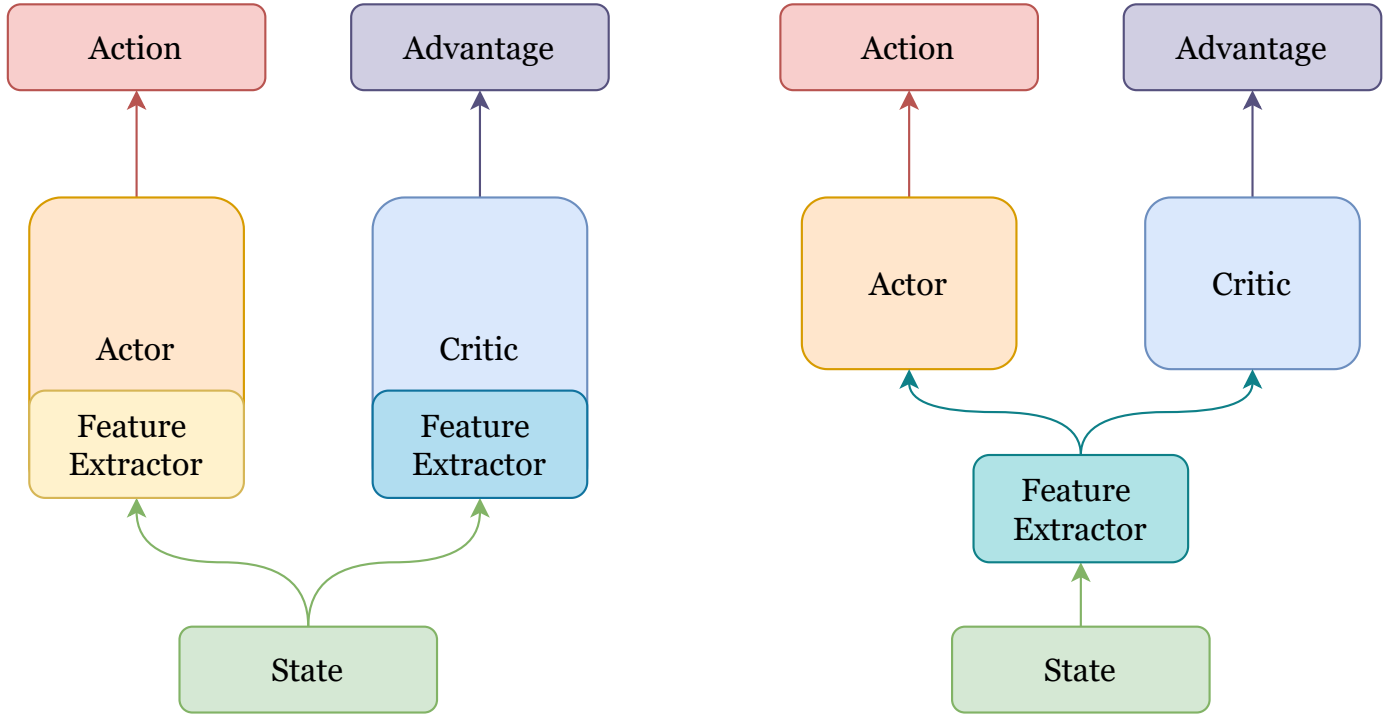
- When the policy is random (i.e., more uniform over actions), H_t is larger
- When the policy becomes deterministic, H_t is smaller.

ADVANTAGE ACTOR-CRITIC (A2C)
REMARKS

- Benefits
  - Denser learning signal → transforms sparse rewards into smoother feedback using the critic
  - Better credit assignment → critic guides the actor on which actions contributed most
  - Lower variance → baseline (value function) reduces update noise
- Drawbacks
  - More complex training → two networks (actor + critic) must be jointly optimized
  - Higher bias → bootstrapping introduces bias compared to pure Monte Carlo
  - Sensitive to hyperparameters → e.g., learning rate balance between actor and critic
  - On-policy limitation → cannot reuse past experiences (less sample efficient)

ADVANTAGE ACTOR-CRITIC (A2C)
REMARKS

- Actor and critic often share the same feature extractor
  - Both depend on state representations
  - This can improve efficiency, stability, and consistency



CODING

TD5: A2C